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	Departamento Department — Stress Methods (TADAP3)	Avión Aircraft	Technologies

Título/Title

PROGRAM KTHOLE USER'S MANUAL

Palabras clave/Key words	Clasificación acceso Access class
MANUALS / HOLES / STRESS CONCENTRATION / COMPUTER PROGRAMS	Airbus Amber **P1**
	Registro de Revisiones Revision Record **Pag. 2**

Resumen/Summary

The purpose of this Technical Report is to define the way to use the KTHOLE program. This program calculates the stress concentration factor Kt for a hole in a panel subjected to a combination of by-pass and transferred axial load. The following effects are considered for the calculation of Kt:

- Net Kt for unloaded hole…………………………Kth
- Bearing Kt for loaded hole………………………Ktbg
- Shadow effect for Kth……………………………Kts
- Thickness effect for total Kt……………………...Ktt
- Countersunk effect for total Kt…………………..Ktc
- Pin bending effect for Ktbg………………………Ktpb

The issue D deals with the release of the version 3.7 of the program.

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F. J. Gomez-Escalonilla Martín (TADAP)		Carlos J. Rodríguez García-Caro	*Electronically validated.*
	Comprobado Checked	Guillermo Moreda González	*Electronically validated.*
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Revisión	Motivo de Modificación / Change reason	Realiz./Prep.	Compr./Check.	Aprobado/App.
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A	Primera edición / First issue	D. Fernández Villalba	L. Claramonte	A. González
September 2003	---	---	---	---
B	Change from version 3.0 to 3.1 and correction of some mistakes found in issue A.	D. Fernández Villalba	---	D. Fernández Villalba
May 2006	Pages 1, 2, 4, 6, 7, 15 and 23.	---	---	---
C	Change from version 3.4 to 3.5. Introduction of Shivkumar's countersunk effect.	J. García Alonso	G. Baños Lapuente	D. Fernández Villalba
October 2015	All pages.	---	---	---
D	Change from version 3.6 to 3.7. Introduction of shared library section	C. Rodriguez	G. Moreda	N/A
March 2025	Section 6 Public Library.	---	---	---

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1 INTRODUCTION

This report makes up the User's Manual for version 3 of the KTHOLE Program, which has been developed by EADS CASA from the year 1993 to date.

This program calculates the stress concentration factor Kt for a hole in a panel subjected to a combination of by-pass and transferred axial load. The following effects are considered for the calculation of Kt:

- Net Kt for unloaded hole, Kth.
- Bearing Kt for loaded hole, Ktbg.
- Shadow effect for Kth, Kts.
- Thickness effect for total Kt, Ktt.
- Countersunk effect for total Kt, Ktc.
- Pin bending effect for Ktbg, Ktpb.

1.1 TECHNICAL SHEET OF THE PROGRAM

Technical Report: NT-T-ADP-03013

PROGRAM	DATE
KTHOLE	March 2025

OBJECTIVE

This program calculates the stress concentration factor Kt for a hole in a panel subjected to a combination of by-pass and transferred axial load.

LANGUAGE	DEVELOPMENT PLATFORM	OS
Fortran 77	Microsoft Visual Studio 10	Windows (x86) and Unix
Visual Basic .Net	Microsoft Visual Studio 10	Windows (x86)

NUMBER OF LINES (APPROX.)	EXECUTION MODE
2200 (Fortran, analysis core) + 1600 (Visual Basic .Net) Total 3800	Console (input data in text file) Interactive GUI

Notes:
- Blank lines and commented lines excluded
- Count for Visual Basic code excludes lines created by Visual Studio in designer.vb files

AUTHOR	VERSION
Dario Fernández Villalba	3.1 to 3.4
Jaime García Alonso	3.5
Maria del Carmen Blanquer Barbera	3.6
Carlos J. Rodríguez García-Caro	3.7

ADDITIONAL DOCUMENTS

See list of applicable references in chapter 1.3.

1.2 ABBREVIATIONS AND SYMBOLS

Symbol	Units	Definition
DLL		Dynamic-link library
GUI		Graphical User Interface
Kt	N/A	Stress Concentration Factor
Kth	N/A	Net Stress Concentration Factor for unloaded hole
Ktbg	N/A	Bearing Stress Concentration Factor for loaded hole
Kts	N/A	Shadow effect for kth
Ktt	N/A	Thickness effect for kth
Ktc	N/A	Countersunk effect for total kt
Ktpb	N/A	Pin bending effect for ktbg
N/A		Not Applicable
OS		Operative System
SO		Shared object

1.3 REFERENCES

	Document Reference	Issue	Title
Ref. 1	NT-T-ADP-03014	B	Theoretical Manual of the Program
Ref. 2	TAE-T-NT-150138	A	Programmer Manual of the Program
Ref. 3	TAE-T-NT-150104	A	Countersunk hole stress concentration

1.4 DIFFERENCES AMONG VERSIONS

<u>Version 1.0:</u>

Initial version developed in Fortran 77 for Digital VAX computer.

<u>Version 2.0:</u>

It is an adaptation of the program to the Sun Solaris computer.

<u>Version 2.1:</u>

Small modification in the calculation of basic Kt's to obtain the same results as from the program AGUJERO.

The following additional effects are included as options:

Shadow effect: The case for intermediate hole has been modified. In version 2.0, this case was considered equal than for end hole.

Thickness effect: This effect has been included in this version.

Countersunk effect: This effect is the same as for the previous version.

Pin bending effect: This effect has been included in this version.

The user can now choose which zone of the hole he wants the results in. These zones are:

Cylindrical zone: The shadow and thickness effects are considered.

Countersunk base: The shadow and countersunk effects are considered.

Surface contact: The shadow and pin bending effects are considered.

The programming code is standardized according to the current practice.

Validation examples are created for further versions of the program.

Version 3.0:

The user can choose the language: Spanish or English.

The program is restructured to allow the creation of a dynamic link library DLL for Windows operative system.

Since the required data are few, the option to enter them by means of an input file is removed.

All the results are presented on the screen. Optionally they can be written in an output file, if required by the user.

Only one example is kept (the output file) for future new versions validation.

Version 3.1:

The GUI (Graphic User Interface) developed in Visual Basic 6.0 is updated to the new development environment of Visual Basic .NET 2003.

The output sheet from the printer is in color or monochrome, depending on the capabilities of the selected device.

Version 3.2:

It is corrected an interpolation error in the rivet’s flexibility effect calculation for $D/t < 0.5$.

Version 3.3:

It is modified the table for rivet’s flexibility effect analysis for $D/t < 0.5$ by the extrapolation the reference curves for this effect. They were defined only for $D/t > 0.5$.

Version 3.4:

The development platform is updated to Microsoft’s Visual Studio 2010 with Intel’s Fortran Parallel Studio XE 2013.

Version 3.5:

It is introduced an additional formulation for countersunk effect, the Shivakumar’s solution. The classic formulation is kept and both are computed, giving as a result the most conservative one. For countersunk depth values higher than 70 % warning messages are produced.

Version 3.6

Several bugs in the program have been fixed:

- In the console version, the program would give an error if S1=S2.

- In the GUI version, the program would not give an error if S1=S2 but P was not zero.

- In the graphical version, if S1=S2, P=0, it would not calculate if the last input was the width (W).

Execution data has been added to the program output. An error has been corrected in the Warning message that warned the user about which formulation was being used to calculate the countersink factor. The message did not correspond to what the program was actually doing.

The "Classic" formulation has been renamed "Northrop."

Version 3.7:

A public shared library has been deployed. The documentation of the subroutine is included in Section 6.

2 PROGRAM EXECUTION

2.1 WINDOWS/PC VERSIONS

The Visual Basic executable program KtHole.exe (as well as an electronic copy of this manual and the program examples) is located in the folder:

```
\\efs\InfoCalculo\Soft\KtHole
```

To execute the program, the user has to click twice on the Visual Basic executable program.

In the subdirectory .\Consola within that folder, there is another executable program with the same name, which is the Fortran program in console mode.

2.2 LINUX/MULTIVAC VERSION

KTHOLE is deployed on the HPC network of Airbus DS at Getafe's facilities. It runs just invoking its name (in lowercase letters) on a command line terminal:

```
kthole
```

Nevertheless, to run this version of the program (console application), you must have already loaded the user profile applicable at the Airbus DS Structural Analysis office, by using the command:

```
. asignaciones_calculo
```

This instruction can be executed in the command terminal prior to execute the program, but it is strongly recommended to include this statement in the files ".profile" and ".kshrc" (if working with Korn Shell) or ".bashrc" (if working with Bash Shell). To do this, the following lines should be included:

```
. asignaciones_calculo
ENV=$HOME/.kshrc
export ENV
BASH_ENV=$HOME/.bashr
export BASH_ENV
```

2.3 SHARED LIBRARY VERSION

A dynamic-link library (.dll file) for Windows and a shared object (.so file) for UNIX-like systems containing official subroutine for this program has been developed, so the users can implement it in their own scripts. For further information see Section 6, where the included subroutines are declared, their input/output arguments are defined, and there is a list of possible errors controlled by the software.

3 INPUT DATA

3.1 Windows or LINUX console applications

The program asks for the language to use, the analysis options and all required data to perform the analysis. The results will be presented on the screen. Additionally, it is possible to specify an output file, as shown in the following example:

```
Idioma / Language:
==================

  1 ... ESPAÑOL

  2 ... ENGLISH

Idioma / Language: 2

.........
.........
.........

Save the results in a file?.................(Y/N): Y
Name of the output results file: Example.txt
```

Details of the interactive data flow on the screen are provided in the example included in this report. The CPU time for calculation is negligible.

3.1.1 Input data

All input data for the program have to be provided in interactive mode. They are enumerated and explained below:

<u>Language:</u>

The user can select Spanish or English in the following screen:

```
Idioma / Language:
==================

  1 ... ESPAÑOL

  2 ... ENGLISH

Idioma / Language: _
```

<u>Options for input data:</u>

Since not all data are independent, it is possible to select which ones will be directly entered and which ones the program will internally calculate, as shown below:

```
            ****************************************************
            *                                                  *
            *   <<< CALCULATION OF Kt's FOR LOADED HOLES >>>   *
            *                                                  *
            ****************************************************

                  +------------------------------------+        +
               <--|  |<--Pitch(1)-->|<--Pitch(2)-->|  |-->      |
               <--|                                    |-->      |
            S1 <--|                  P                 |--> S2   W
               <--|  O              O====>          O  |-->      |   +
               <--|                                    |-->      |   ED
                  +------------------------------------+        +   +

Options for input data:
=======================

Stresses: S2 < Smed < S1      Smed=(S1+S2)/2
Fastener load: P              Panel width: W

   1 ..... Data: S1   - S2 - P

   2 ..... Data: Smed - P  - W

Select the desired option: 1
```

The variables to be entered and calculated in each case are the following:

- Option1:
  - Entered:

    S_1 Far field stress before load transfer.

    S_2 Far field stress after load transfer.

    P Transferred load.

    t Plate thickness.

  - Calculated:

    S_{med} Mean stress.

$$S_{med} = \frac{S_1 + S_2}{2}$$

    W Plate width.

$$W = \frac{P}{t \cdot (S_1 - S_2)}$$

- Option2:
  - Entered:

    S_{med} Mean stresses.

    P Transferred load.

    W Plate width.

    t Plate thickness.

  - Calculated:

    S_1 Far field stress before load transfer.

$$S_1 = S_{med} + \frac{P}{2 \cdot W \cdot t}$$

    S_2 Far field stress after load transfer.

$$S_2 = S_{med} - \frac{P}{2 \cdot W \cdot t}$$

<u>Options for analysis:</u>

The following effects are optional in the calculation of the stress concentration factor Kt:

- Shadow effect.
- Thickness effect.
- Countersunk effect. The program selects automatically the most conservative option for countersunk effect between Classic effect and Shivakumar's. For further information see the theoretical manual of the KtHole program.
- Pin bending effect.

The user can select the above options in the following screen:

```
Options for analysis:
=====================

   - Shadow effect......................(Y/N): Y

   - Thickness effect...................(Y/N): Y

   - Countersunk effect.................(Y/N): Y

   - Pin bending effect.................(Y/N): Y
```

Type of joint:

The user can select the type of joint for the element to be analyzed, in the following screen:

```
Type of joint:
==============

   1 ..... Single shear

   2 ..... Double shear

Select the type of joint: 1
```

The option 1 must be selected when the element to be analyzed is in a single shear joint or it is an external plate in a double shear joint:

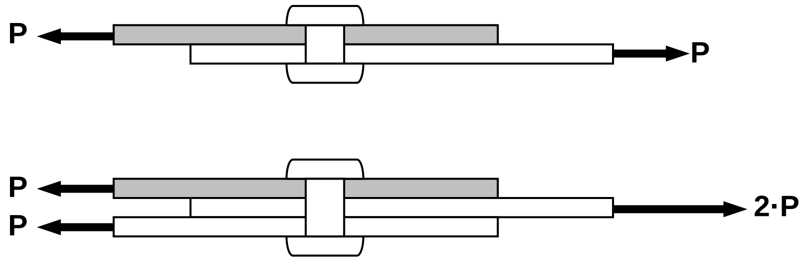

Figure 1. Elements in single shear

The option 2 must be selected when the element to be analyzed is the central one in a double shear joint:

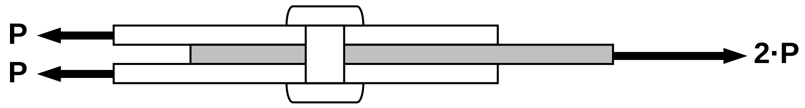

Figure 2. Elements in double shear

If the pin bending effect has not been selected, the type of joint is not relevant for the Kt calculation and the above screen will not be presented.

Problem data:

All data defining the problem have to be entered by means of the following screen, with exception of those variables that the program will calculate automatically, according to the selected "Option for input data":

```
Problem data:
=============

   Stress at section 1.............S1 [    0.00]: 100
   Stress at section 2.............S2 [    0.00]: 50
   Mean stress.....................Sm [    0.00]: 75
   Fastener load...................P  [     0.0]: 2500
   Panel width.....................W  [    0.00]: 25
   Distance from hole to edge......ED [    0.00]: 10
   Panel thickness.................t  [    0.00]: 2
   Hole diameter...................D  [    0.00]: 4
   Fasteners pitch in 1............P1 [    0.00]: 20
   Fasteners pitch in 2............P2 [    0.00]: 30
   Countersunk depth...............c  [    0.00]: 1
   Equivalent thickness...........teq[    0.00]: 1
   Fastener Young's modulus........Er [      0.]: 110000
   Panel Young's modulus...........Ep [      0.]: 71000
```

The data between brackets are the default values (taken by the program if the `Enter` key is pressed without any value). Initially, all default data are set to zero. If the user wants to change any value after the program execution, the above screen will be presented, but with the current data between brackets as default values.

- The program will not require some of the above data. It depends on the previously selected options, as follows:
- Stress before load transfer, S1: Required if the option for input data is 1 (S1 – S2 – P).
- Stress after load transfer, S2: Required if the option for input data is 1 (S1 – S2 – P).
- Mean stress, Sm: Required if the option for input data is 2 (Smed – P – W).
- Panel width, W: Required if the option for input data is 2 (Smed – P – W) or option for input data is 1 (S1 – S2 – P), P=0 and S1=S2.
- Countersunk depth, c: Required if countersunk effect has been selected.
- Equivalent thickness, teq: Required if pin bending effect has been selected and the considered element is in double shear.
- Young's modulus for fastener and panel, Er and Ep: Required if pin bending effect has been selected.

The equivalent thickness to be specified in case of central element in a double shear joint must be defined as:

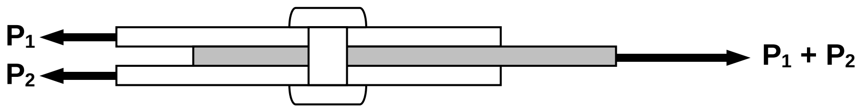

Figure 3. Double shear joint

$$t_{eq}=\frac{P_1}{P_1+P_2}$$ Considering the upper surface contact

$$t_{eq}=\frac{P_2}{P_1+P_2}\cdot t$$ Considering the lower surface contact

In a general case where the relationship between loads P_1 and P_2 is unknown, it is recommended to use:

$$t_{eq}=\frac{t}{2}$$

3.2 Windows application

An application has been developed in Visual Basic for Windows environment. The original routines of the program for calculation and output results have been included in a Dynamic Link Library, which is used by this application. Therefore, the same results will be obtained, with only differences due to rounding.

Since this is a Windows application, it must be installed in the user's PC before it can be executed. For installation, please consult with the system manager.

The input data must be typed in the application form. When all data have been introduced, the results are automatically presented.

The form, after loading the data corresponding to the example (see section 7), has the following appearance:

The example data can be loaded by double click on the first frame picture.

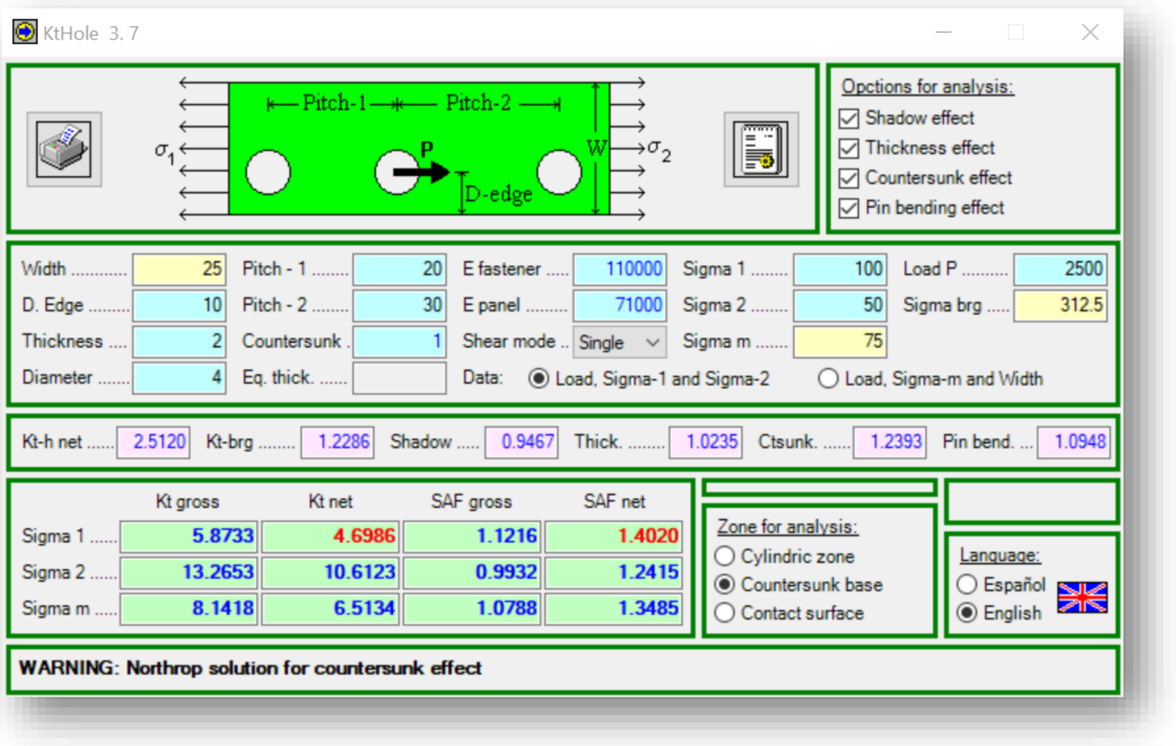

Figure 1. KtHole windows application input data form

The buttons located at both sides of the picture are to write the results in a printer (left button) or to write the results in a file (right button).

3.3 Shared library

Input/Output arguments of dynamic-link library (.dll or .so file) are defined in Section 6.

4 OUTPUT RESULTS

The KTHOLE program provides the analysis results on the screen, in interactive way. After the input data, the following results are displayed:

Input data:

All input data are displayed on the screen, as shown below:

```
          **************************************************
          *                                                *
          *   <<< CALCULATION OF Kt FOR LOADED HOLES >>>   *
          *                                                *
          **************************************************

                +------------------------------------+        +
             <--|  |<---- P1 ---->|<---- P2 ---->|   |-->     |
             <--|                                    |-->     |
          S1 <--|                  P                 |--> S2  W
             <--|  O             O====>          O   |-->     |  +
             <--|                                    |-->     |  ED
                +------------------------------------+        +  +

Options for analysis:
=====================
Shadow effect....................... Yes
Thickness effect.................... Yes
Countersunk effect.................. Yes (Classic)
Pin bending effect.................. Yes

Press <Enter> to continue _



Input data:
===========
Panel width.....................W   =     25.00
Distance from hole to edge......ED  =     10.00
Panel thickness.................t   =      2.00
Hole diameter...................D   =      4.00
Fasteners pitch in 1............P1  =     20.00
Fasteners pitch in 2............P2  =     30.00
Countersunk depth...............c   =      1.00
Type of joint....................... Single shear
Fastener Young's modulus........Er  = 110000.00
Panel Young's modulus...........Ep  =  71000.00
Stress at section 1.............S1  =    100.00
Stress at section 2.............S2  =     50.00
Mean stress.....................Sm  =     75.00
Fastener load...................P   =   2500.00
Bearing stress..................Sbg =    312.50

Press <Enter> to continue _
```

Basic Kt’s:

Displayed on the following screen:

```
Basic Kt's:
===========
Net Kt for unloaded hole..............Kth  =   2.5120
Bearing Kt for loaded hole............Ktbg =   1.2286
Shadow effect for Kth.................Kts  =   0.9467
Thickness effect for total Kt.........Ktt  =   1.0235
Countersunk effect for total Kt.......Ktc  =   1.2393
Pin bending effect for Ktbg...........Ktpb =   1.0948

Press <Enter> to continue _
```

Final results:

Displayed as follows for the three possible locations:

```
Results for the cylindrical zone:
=================================
          Kt gross        Kt net            SAF gross       SAF net
         ----------     ----------         -----------     -----------
S1 --->     4.8506         3.8804            1.121587        1.401983
S2 --->    10.9554         8.7643            0.993173        1.241467
Sm --->     6.7240         5.3792            1.078782        1.348478

Results for the countersunk base:
=================================
          Kt gross        Kt net            SAF gross       SAF net
         ----------     ----------         -----------     -----------
S1 --->     5.8733         4.6986            1.121587        1.401983
S2 --->    13.2653        10.6123            0.993173        1.241467
Sm --->     8.1418         6.5134            1.078782        1.348478

Results for the contact surface:
================================
          Kt gross        Kt net            SAF gross       SAF net
         ----------     ----------         -----------     -----------
S1 --->     5.0639         4.0511            1.121587        1.401983
S2 --->    11.4372         9.1498            0.993173        1.241467
Sm --->     7.0197         5.6158            1.078782        1.348478
```

Results for the countersunk base are only provided if the countersunk effect has been included in the analysis.

In the same way, the results for the surface contact are displayed if the pin bending effect has been selected.

Output file:

It is possible to keep all the above information in a text file. For this purpose, the program asks if the user wants to save the results in a file. The last step is to ask for the modification of the input data, showing the following screen:

```
Save the results in a file?.................(Y/N): Y
Name of the output results file: Example.txt

Do you want to modify any input data?.......(Y/N): N



KtHole 3.7. (c) Airbus Defence and Space 2025
```

If the answer to the possible modifications is N, the program ends the execution. Otherwise, the screen for input data would be presented. In this case, the default problem data, presented between brackets [ ], would be the current data.

Warning messages:

When the program detects some special situation concerning the processing of data it produces a warning message for information purposes.

All warnings are related with the adopted solution for countersunk evaluation depending on the hole parameters and the countersunk depth limitations.

Warning messages. English output:

```
WARNING: Classic solution for countersunk effect
WARNING: Shivakumar´s solution for countersunk effect
WARNING: Classic solution for countersunk effect. Countersunk depth out of
         design criteria c/t < 0.7
WARNING: Shivakumar´s solution for countersunk effect. Countersunk depth
         out of design criteria c/t < 0.7
```

Warning messages. Spanish output:

```
AVISO: Solución clásica para el efecto de avellanado
AVISO: Solución de Shivakumar para el efecto de avellanado
AVISO: Solución clásica para efecto de avellanado. Avellanado fuera de los
       criterios de diseño:  c/t<0.7
AVISO: Solución de Shivakumar para efecto de avellanado. Avellanado fuera
       de los criterios de diseño:  c/t<0.7
```

Error messages:

When the program detects an error in the input data, it writes the corresponding message in the console and repeats the whole input data process. The error messages that might appear in the console are listed below:

Error messages. English output:

```
ERROR: The stress at section 1 must be greater than zero
ERROR: The stress at section 2 must be greater or equal than zero
```

```
ERROR: The stress at section 2 must be greater or equal than in section 2
ERROR: The fastener load cannot be negative
ERROR: The fastener load must be greater than zero
ERROR: The panel width must be greater than zero
ERROR: The distance from hole to edge must be greater than zero
ERROR: The panel width must be greater than twice the distance from hole
       to edge
ERROR: The panel thickness must be greater than zero
ERROR: The hole diameter must be greater than zero
ERROR: The distance from hole to edge must be greater than the radius
ERROR: The fasteners pitch at section 1 must be greater than the diameter
ERROR: The fasteners pitch at section 2 must be greater than the diameter
ERROR: The countersunk depth must be: 0 < c < t
ERROR: The equivalent thickness of a panel in double shear must be:
       0 < teq < t
ERROR: The fastener Young's modulus must be positive
ERROR: The panel Young's modulus must be positive
ERROR: The output file cannot be opened for writing
ERROR: It is not possible to write in the output file
ERROR: The load P must be 0 if S1=S2
```

Error messages. Spanish output:

```
ERROR: El esfuerzo en la sección 1 debe ser mayor que cero
ERROR: El esfuerzo en la sección 2 debe ser mayor o igual que cero
ERROR: El esfuerzo en la sección 1 debe ser mayor o igual que en la
       sección 2
ERROR: La carga en el remache no puede ser negativa
ERROR: La carga en el remache debe ser mayor que cero
ERROR: La anchura del panel debe ser mayor que cero
ERROR: La distancia del agujero al borde debe ser mayor que cero
ERROR: La anchura del panel debe ser mayor de dos veces la distancia del
       agujero al borde
ERROR: El espesor del panel debe ser mayor que cero
ERROR: El diámetro del agujero debe ser mayor que cero
ERROR: La distancia del agujero al borde debe ser mayor que el radio
ERROR: El paso entre remaches en la sección 1 de be ser mayor que el
       diámetro
ERROR: El paso entre remaches en la sección 2 de be ser mayor que el
       diámetro
ERROR: La profundidad de avellanado debe ser: 0 < c < t
ERROR: El espesor equivalente de un panel en doble cortadura debe ser:
       0 < teq < t
ERROR: El módulo de elasticidad del remache debe ser positivo
ERROR: El módulo de elasticidad del panel debe ser positivo
ERROR: No se puede abrir el fichero de salida para escritura
ERROR: No se puede escribir en el fichero de salida
ERROR: La carga P debe ser 0 con S1 = S2
```

The final results have the following meaning:

- **Kt gross:** Gross stress concentration factor. Is the relationship between the maximum stress in the hole contour and the equivalent gross stress in the symmetric case.
- **Kt net:** Net stress concentration factor. Is the relationship between the maximum stress in the hole contour and the equivalent net stress in the symmetric case.
- **SAF gross:** Stress adjustment factor for gross stress. Is the relationship between the equivalent gross stress and the average gross stress.
- **SAF net:** Stress adjustment factor for net stress. Is the relationship between the equivalent net stress and the average gross stress.

The maximum stress in the contour σ_{max} for each zone of analysis can be evaluated in six different ways. They are:

$$\begin{aligned}\sigma_{max} &= S_1 \cdot SAF_{gross}(S_1) \cdot Kt_{gross}(S_1) = S_1 \cdot SAF_{net}(S_1) \cdot Kt_{net}(S_1) = \\ &= S_2 \cdot SAF_{gross}(S_2) \cdot Kt_{gross}(S_2) = S_2 \cdot SAF_{net}(S_2) \cdot Kt_{net}(S_2) = \\ &= S_m \cdot SAF_{gross}(S_m) \cdot Kt_{gross}(S_m) = S_m \cdot SAF_{net}(S_m) \cdot Kt_{net}(S_m)\end{aligned}$$

For fatigue analysis purposes it is conservatively recommended to use:

- Average gross stress: S_1
- Stress concentration factor Kt: $Kt_{net}(S_1)$
- Stress adjustment factor SAF: $SAF_{net}(S_1)$

5 LIMITATIONS

The methods for obtaining the stress concentration factor for countersunk depths exceeding the 70% of the thickness are supported by numerical FEM analyses performed by various authors and in-house validation, but there is not a universal agreement about the robustness of the approach, as not contrasted with a vast test and service experience achieved in designs following the standard practice. That’s why the use of this tool for countersunk depths over the 70% limit shall be explicitly authorized by the local stress authority of the project. Further information can be found in Ref. 3.

6 PUBLIC LIBRARY

In order to allow the use of this official analysis tool within practical software created by users, a public library has been created. Thus, the user will be able to import official subroutines to their practical software, carry out the analysis with this official software and use the results within their scripts. Therefore, those subroutines will only receive the inputs and will return the results via computer's memory, without writing any external file. These methods shall be progressively made accessible to external software by means of a dynamic link library DLL built for Windows applications and shared object SO for Unix-like systems.

The Windows shared libraries have been compiled with the Intel Fortran compiler (ifort 2018) for Windows 64bits architecture and 32bits architecture, using the compile option `/iface:mixed_str_len_arg` for defining the interface convention for passing character strings, and the following compiler directive on the entry points (subroutine):

```
!DEC$ ATTRIBUTES DLLEXPORT,STDCALL :: name_of_entry-point_subroutine
```

Notes:

- The arguments of the entry points (subroutines) are of interoperable C types (in module `iso_c_binding`).
- It is required to use the same variable type (and dimension) in the entry-point subroutine and in the interface subroutine. The specific features of the input/output arguments for each entry-point subroutine are provided later on. In the case of array arguments note that, in addition to the argument, it is required to pass the dimension.
- To invocate the DLLs/SOs from Python scripts, a demo code is provided to help the user with the calling conventions
- To invocate from Visual Basic scripts in MS Excel, it is required to use the x32 bits library file.

Pre-requisites

The computer and user must comply with the following pre-requisites to be able to use the shared library:

- Microsoft Windows 10 (64 bits) Operating System.

- Redistributable libraries of Intel® Fortran Compiler for Windows, which can be installed via Automatic Remote Software Installations (ARSI) from the Airbus DS HPC.

- Environment variable (at user or system level) in your PC, with name BASES_DATOS, its value assigned to the path where the license file (and other databases for the Stress Software) will be copied, ended by a backslash character.

- Licence file (at BASES_DATOS path) with user authenticated by the Stress Methods and Optimization function of Airbus DS

KTHOLE ANALYSIS:

Performs a complete KT hole analysis calculating all possible values. Please, check the output values in the table of the arguments below. If the effect is not selected, the corresponding kt value will be 1.0 (e.g., if eff_shadow is 0 (False), the value in kt_shadow will be 1.0)

Name of the entry-point subroutine: ***analysis_kthole***

List of arguments:

#	*Name*	*Type*	*Dim*	*Bytes*	*Intent*	*Description*
1	width	REAL (c_float)	-	8	IN	Panel width (W)
2	edge_dist	REAL (c_float)	-	8	IN	Distance from hole to edge (ED)
3	thk	REAL (c_float)	-	8	IN	Panel thickness (t)
4	diam	REAL (c_float)	-	8	IN	Hole diameter (D)
5	xpitch1	REAL (c_float)	-	8	IN	Fasteners pitch in 1 (P1)
6	xpitch2	REAL (c_float)	-	8	IN	Fasteners pitch in 2 (P2)
7	csnk_depth	REAL (c_float)	-	8	IN	Countersunk depth (c)
8	thk_eq	REAL (c_float)	-	8	IN	Equivalent thickness to be specified in case of central element in a double shear joint (teq)
9	E_fastener	REAL (c_float)	-	8	IN	Fastener Young's modulus (Er)
10	E_panel	REAL (c_float)	-	8	IN	Panel Young's modulus (Ep)
11	sigma1	REAL (c_float)	-	8	IN	Stress at section 1 (S1)
12	sigma2	REAL (c_float)	-	8	IN	Stress at section 2 (S2)
13	sigmamed	REAL (c_float)	-	8	IN	Mean stress (Sm)
14	P	REAL (c_float)	-	8	IN	Fastener load (P)
15	mode_loads	INTEGER (c_int)	-	4	IN	Input load selected: 1 for S_1-S_2-P, 2 for S_{med}-P-W
16	mode_shear	INTEGER (c_int)	-	4	IN	Type of joint (1 for single shear, 2 for double shear)
17	eff_shadow	INTEGER (c_int)	-	4	IN	Shadow effect option (1 for True, 0 for False)
18	eff_thk	INTEGER (c_int)	-	4	IN	Thickness effect option (1 for True, 0 for False)
19	eff_csnk	INTEGER (c_int)	-	4	IN	Countersunk effect option (1 for True, 0 for False)
20	eff_pinbnd	INTEGER (c_int)	-	4	IN	Pin bending effect option (1 for True, 0 for False)
21	kt_net	REAL (c_float)	-	8	OUT	Net Kt for unloaded hole (Kth)
22	kt_brg	REAL (c_float)	-	8	OUT	Bearing Kt for loaded hole (Ktbg)

23	kt_shadow	REAL (c_float)	-	8	OUT	Shadow effect for Kth (Kts)
24	kt_thk	REAL (c_float)	-	8	OUT	Thickness effect for total Kt (Ktt)
25	kt_csnk	REAL (c_float)	-	8	OUT	Countersunk effect for total Kt (Ktc)
26	kt_pinbnd	REAL (c_float)	-	8	OUT	Pin bending effect for Ktbg (Ktpb)
27	kt_gross_s1	REAL (c_float)	3	8*3=24	OUT	Kt gross at section 1 (first element for cylindrical zone, second element for countersunk base and third element for contact surface)
28	kt_net_s1	REAL (c_float)	3	8*3=24	OUT	Kt net at section 1 (first element for cylindrical zone, second element for countersunk base and third element for contact surface)
29	kt_gross_s2	REAL (c_float)	3	8*3=24	OUT	Kt gross at section 2 (first element for cylindrical zone, second element for countersunk base and third element for contact surface)
30	kt_net_s2	REAL (c_float)	3	8*3=24	OUT	Kt net at section 2 (first element for cylindrical zone, second element for countersunk base and third element for contact surface)
31	kt_gross_sm	REAL (c_float)	3	8*3=24	OUT	Kt gross for mean stress (first element for cylindrical zone, second element for countersunk base and third element for contact surface)
32	kt_net_sm	REAL (c_float)	3	8*3=24	OUT	Kt gross for mean stress (first element for cylindrical zone, second element for countersunk base and third element for contact surface)
33	safg_s1	REAL (c_float)	-	8	OUT	Gross Stress Adjustment Factor at section 1
34	safn_s1	REAL (c_float)	-	8	OUT	Net Stress Adjustment Factor at section 1
35	safg_s2	REAL (c_float)	-	8	OUT	Gross Stress Adjustment Factor at section 2
36	safn_s2	REAL (c_float)	-	8	OUT	Net Stress Adjustment Factor at section 2
37	safg_sm	REAL (c_float)	-	8	OUT	Gross Stress Adjustment Factor for mean stress
38	safn_sm	REAL (c_float)	-	8	OUT	Net Stress Adjustment Factor for mean stress
39	header	CHARACTER (c_char)	21	21	OUT	Character string containing the name of the stress method and its version
40	error_code	INTEGER (c_int)	-	4	OUT	Error code: 0 is success; for other values, check next table *Error codes*.
41	warn_code	INTEGER (c_int)	-	4	OUT	Warning code related with the solution for countersunk evaluation

Table 1. List of arguments for analysis_kthole subroutine

Error codes:

Error code	***Description***
101	The stress at section 1 must be greater than zero
102	The stress at section 2 must be greater or equal than zero
103	The stress at section 1 must be greater or equal than in section 2
104	The fastener load cannot be negative
105	The fastener load must be greater than zero
106	The panel width must be greater than zero
107	The distance from hole to edge must be greater than zero
108	The panel width must be greater than twice the distance from hole to edge
109	The panel thickness must be greater than zero
110	The hole diameter must be greater than zero
111	The distance from hole to edge must be greater than the radius
112	The fasteners pitch at section 1 must be greater than the diameter
113	The fasteners pitch at section 2 must be greater than the diameter
114	The countersunk depth must be: $0 < c < t$
115	The equivalent thickness of a panel in double shear must be: $0 < teq < t$
116	The fastener Young's modulus must be positive
117	The panel Young's modulus must be positive
120	The load P must be 0 if S1=S2
900	You are not authorized to use this program
901	Licence file not found
905	The licence file cannot be opened
906	The licence file cannot be read
907	The licence file cannot be read
908	You are not authorised to use this program (User does not exist / Licence expired)

Table 2. Error codes for analysis_kthole subroutine

7 EXAMPLES

The case to be solved consists in an axial joint between two plates, fastened each other, as shown in the following figure:

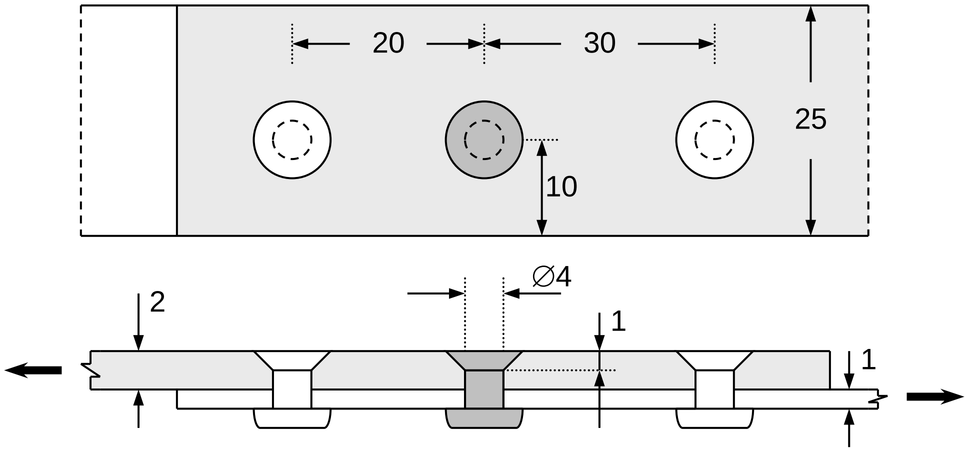

Figure 2. Example geometry.

The hole under study will be the central one. The load transferred through the fastener as well as the average stresses at both sides of the hole must be determined by means of a load transfer analysis (i.e. running the ARES program). In this example, the following stresses will be assumed:

- Average stress on the left side of the central hole, S1 = 100 N/mm2
- Average stress on the right side of the central hole, S2 = 50 N/mm2

The load transferred through the fastener can be then determined by applying the equilibrium of forces at both sides of the hole, resulting:

- Transferred load, $P = (S_1 - S_2) \cdot W$ = 2500 N

All effects on Kt will be accounted for. These effects are:

- Shadow effect, Yes.
- Thickness effect, Yes.
- Countersunk effect, Yes.
- Pin bending effect, Yes.

With all the above data, now it is possible to run the program. The screen data flow can be the following, starting by the language selection:

```
Idioma / Language:
==================

  1 ... ESPAÑOL

  2 ... ENGLISH

Idioma / Language: 2
```

The second question is the option for input data. In this case, the option 1 has been selected because S_1, S_2 and P have been previously defined:

```
            *****************************************************
            *                                                   *
            *   <<< CALCULATION OF Kt's FOR LOADED HOLES >>>    *
            *                                                   *
            *****************************************************

                  +------------------------------------+        +
               <--|  |<--Pitch(1)-->|<--Pitch(2)-->|   |-->     |
               <--|                                    |-->     |
            S1 <--|                  P                 |--> S2  W
               <--|  O            O====>          O    |-->     |  +
               <--|                                    |-->     |  ED
                  +------------------------------------+        +  +

Options for input data:
=======================

Stresses: S2 < Smed < S1     Smed=(S1+S2)/2
Fastener load: P             Panel width: W

   1 ..... Data: S1   - S2 - P

   2 ..... Data: Smed - P  - W

Select the desired option: 1
```

As pointed above, all possible effects on Kt will be accounted for and the plate it is in a single shear joint. Therefore:

```
Options for analysis:
=====================

   - Shadow effect......................(Y/N): Y

   - Thickness effect...................(Y/N): Y

   - Countersunk effect.................(Y/N): Y

   - Pin bending effect.................(Y/N): Y

Type of joint:
==============

   1 ..... Single shear

   2 ..... Double shear

Select the type of joint: 1
```

The next step is the introduction of the geometry data and the load case:

```
Problem data:
=============

   Stress at section 1.............S1 [    0.00]: 100
   Stress at section 2.............S2 [    0.00]: 50
   Mean stress.....................Sm [    0.00]: 75
   Fastener load...................P  [     0.0]: 2500
   Panel width.....................W  [    0.00]: 25
   Distance from hole to edge......ED [    0.00]: 10
   Panel thickness.................t  [    0.00]: 2
   Hole diameter...................D  [    0.00]: 4
   Fasteners pitch in 1............P1 [    0.00]: 20
   Fasteners pitch in 2............P2 [    0.00]: 30
   Countersunk depth...............c  [    0.00]: 1
   Equivalent thickness............teq[    0.00]: 1
   Fastener Young's modulus........Er [      0.]: 110000
   Panel Young's modulus...........Ep [      0.]: 71000
```

In this point, the program solves the problem and displays the following results, starting by an echo of the input data. First, the options of analysis are printed:

```
            **************************************************
            *                                                *
            *   <<< CALCULATION OF Kt FOR LOADED HOLES >>>   *
            *                                                *
            **************************************************

                  +------------------------------------+        +
               <--|  |<---- P1 ---->|<---- P2 ---->|   |-->      |
               <--|                                    |-->      |
            S1 <--|                  P                 |--> S2   W
               <--|  O             O===>          O    |-->      |  +
               <--|                                    |-->      |  ED
                  +------------------------------------+        +  +

Options for analysis:
=====================
Shadow effect....................... Yes
Thickness effect.................... Yes
Countersunk effect.................. Yes (Classic)
Pin bending effect.................. Yes

Press <Enter> to continue _
```

The next screen is the input data, including those data calculated by the program. In this case they are the panel width W, the mean stress Sm and the bearing stress Sbg:

```
Input data:
===========
Panel width.....................W   =      25.00
Distance from hole to edge......ED  =      10.00
Panel thickness.................t   =       2.00
Hole diameter...................D   =       4.00
Fasteners pitch in 1............P1  =      20.00
Fasteners pitch in 2............P2  =      30.00
Countersunk depth...............c   =       1.00
Type of joint....................... Single shear
Fastener Young's modulus........Er  =  110000.00
Panel Young's modulus...........Ep  =   71000.00
Stress at section 1.............S1  =     100.00
Stress at section 2.............S2  =      50.00
Mean stress.....................Sm  =      75.00
Fastener load...................P   =    2500.00
Bearing stress..................Sbg =     312.50

Press <Enter> to continue _
```

Next screens show the basic Kt’s, including the selected effects, and the final results:

```
Basic Kt's:
===========
Net Kt for unloaded hole..............Kth  =   2.5120
Bearing Kt for loaded hole............Ktbg =   1.2286
Shadow effect for Kth.................Kts  =   0.9467
Thickness effect for total Kt.........Ktt  =   1.0235
Countersunk effect for total Kt.......Ktc  =   1.2393
Pin bending effect for Ktbg...........Ktpb =   1.0948

Press <Enter> to continue _

Results for the cylindrical zone:
=================================
          Kt gross      Kt net            SAF gross      SAF net
         ----------    ----------        -----------    -----------
S1 --->     4.8506        3.8804           1.121587       1.401983
S2 --->    10.9554        8.7643           0.993173       1.241467
Sm --->     6.7240        5.3792           1.078782       1.348478

Results for the countersunk base:
=================================
          Kt gross      Kt net            SAF gross      SAF net
         ----------    ----------        -----------    -----------
S1 --->     5.8733        4.6986           1.121587       1.401983
S2 --->    13.2653       10.6123           0.993173       1.241467
Sm --->     8.1418        6.5134           1.078782       1.348478

Results for the contact surface:
================================
          Kt gross      Kt net            SAF gross      SAF net
         ----------    ----------        -----------    -----------
S1 --->     5.0639        4.0511         1.121587       1.401983
S2 --->    11.4372        9.1498         0.993173       1.241467
Sm --->     7.0197        5.6158         1.078782       1.348478
```

Finally, the program asks to the user for saving the results in a file and for modifications in the input data. In this case, results will be kept in the file “Example.txt” and no modification will be introduced. Then:

```
Save the results in a file?.................(Y/N): Y
Name of the output results file: Example.txt

Do you want to modify any input data?.......(Y/N): N



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```

The output file is listed below:

Output results file:

```
*************************************************
*                                               *
*   <<< CALCULATION OF Kt FOR LOADED HOLES >>>  *
*                                               *
*************************************************

      +------------------------------------+        +
   <--|  |<---- P1 ---->|<---- P2 ---->|  |-->      |
   <--|                                   |-->      |
S1 <--|                 P                 |--> S2   W
   <--|  O             O====>         O   |-->      |  +
   <--|                                   |-->      |  ED
      +------------------------------------+        +  +

Options for analysis:
=====================
Shadow effect....................... Yes
Thickness effect.................... Yes
Countersunk effect.................. Yes (Classic)
Pin bending effect.................. Yes

Input data:
===========
Panel width.....................W   =      25.00
Distance from hole to edge......ED  =      10.00
Panel thickness.................t   =       2.00
Hole diameter...................D   =       4.00
Fasteners pitch in 1............P1  =      20.00
Fasteners pitch in 2............P2  =      30.00
Countersunk depth...............c   =       1.00
Type of joint....................... Single shear
Fastener Young's modulus........Er  =  110000.00
Panel Young's modulus...........Ep  =   71000.00
Stress at section 1.............S1  =     100.00
Stress at section 2.............S2  =      50.00
Mean stress.....................Sm  =      75.00
Fastener load...................P   =    2500.00
Bearing stress..................Sbg =     312.50
Output results file: (cont.)

Basic Kt's:
===========
Net Kt for unloaded hole...............Kth  =   2.5120
Bearing Kt for loaded hole.............Ktbg =   1.2286
Shadow effect for Kth..................Kts  =   0.9467
Thickness effect for total Kt..........Ktt  =   1.0235
Countersunk effect for total Kt........Ktc  =   1.2393
Pin bending effect for Ktbg............Ktpb =   1.0948

Results for the cylindrical zone:
```

```
=================================
          Kt gross       Kt net             SAF gross       SAF net
         ----------     ----------         -----------     -----------
S1 --->     4.8506        3.8804             1.121587        1.401983
S2 --->    10.9554        8.7643             0.993173        1.241467
Sm --->     6.7240        5.3792             1.078782        1.348478

Results for the countersunk base:
=================================
          Kt gross       Kt net             SAF gross       SAF net
         ----------     ----------         -----------     -----------
S1 --->     5.8733        4.6986             1.121587        1.401983
S2 --->    13.2653       10.6123             0.993173        1.241467
Sm --->     8.1418        6.5134             1.078782        1.348478

Results for the contact surface:
================================
          Kt gross       Kt net             SAF gross       SAF net
         ----------     ----------         -----------     -----------
S1 --->     5.0639        4.0511             1.121587        1.401983
S2 --->    11.4372        9.1498             0.993173        1.241467
Sm --->     7.0197        5.6158             1.078782        1.348478

Program KtHole. Version 3.7. (c) Airbus Defence and Space 2025.
```